

NANDU PANAKANTI

AI Engineer · Full-Stack Developer · Agent & RAG Systems Builder

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PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Developer with hands-on experience shipping production-ready AI systems - agent orchestration, RAG pipelines, LLM APIs, and event-driven backends. Evaluated on execution, deployment quality, and real-world usability. Experienced across FastAPI, Node.js, React, PostgreSQL, and AWS. Brings end-to-end ownership from architecture to live deployment.

TECHNICAL SKILLS

AI & LLM: LLM APIs (OpenAI, Anthropic), Agent Orchestration, RAG & Retrieval, Prompt Engineering, Structured Outputs, Tool Calling, AI Workflow Automation

Backend: FastAPI, Node.js, Express.js, REST APIs, Microservices, BullMQ, Redis, WebSockets

Frontend: React.js, TypeScript, JavaScript ES6, HTML5, CSS3, Responsive UI

Databases: PostgreSQL, MongoDB, Supabase, MySQL, SQLAlchemy, Query Optimization

Cloud & DevOps: AWS EC2/S3/Lambda, Docker, GitHub Actions, CI/CD Pipelines, Render

Security: JWT Auth, RBAC, CSRF Protection, OAuth 2.0, API Security, Rate Limiting

SELECTED PROJECTS

PropMind - AI-Powered Property Management Platform

Apr 2026 - May 2026

Node.js · Express.js · MongoDB · OpenAI API · Stripe · BullMQ · Redis · Socket.io [GitHub](#) [Admin Demo](#) [Tenant Demo](#)

- Architected a full-stack AI-driven property management platform automating the complete landlord-tenant lifecycle: applications, leasing, Stripe payments, and maintenance.
- Integrated OpenAI for AI-powered tenant risk scoring, maintenance ticket triage with priority classification, and a natural-language assistant querying live database data.
- Built event-driven payment workflows with Stripe Checkout, webhook processing, and a double-entry ledger system; automated billing via BullMQ/Redis (invoice generation, late fees, rent reminders).
- Deployed dual-portal monorepo on Render with CI/CD, JWT auth, CSRF protection, rate limiting, and real-time notifications via Socket.io: production-grade security end to end.

AI SaaS Operations Copilot

May 2026 - Jun 2026

FastAPI · PostgreSQL · SQLAlchemy · React · TypeScript · GitHub API · OAuth 2.0 [GitHub](#) [Live Demo](#)

- Engineered an AI-driven operational intelligence platform scanning GitHub repositories to generate actionable recommendations for security, cost optimization, and repo governance.
- Built end-to-end GitHub OAuth flow, secure session management, and repo sync workflows; developed analytics modules for dormant repo detection, health monitoring, and CI/CD waste estimation.
- Integrated LLM-powered recommendation engine and conversational AI Copilot allowing natural-language queries over repository risk, cost, and maintenance data.
- Designed scalable FastAPI + SQLAlchemy + PostgreSQL backend with modular service and analytics layers; deployed on Vercel with full TypeScript React frontend.

AI Workflow Automation Platform

Mar 2026 - Apr 2026

FastAPI · PostgreSQL · React · AWS EC2/S3 · Docker · GitHub Actions [GitHub](#) [Live Demo](#)

- Designed and deployed a full-stack AI workflow automation platform with LLM API integration, prompt engineering, structured outputs, and orchestrated task execution.
- Built React frontend and FastAPI backend with PostgreSQL schemas optimized for workflow storage and analytics; deployed on AWS EC2/S3 with Docker and GitHub Actions CI/CD.

Employee Attrition Prediction System

Oct 2025 - Nov 2025

Python · Scikit-learn · SHAP · Streamlit [GitHub](#) [Live Demo](#)

- Built end-to-end ML pipeline for attrition prediction with explainable AI via SHAP, automated preprocessing, feature engineering, and an interactive Streamlit evaluation dashboard.

PROFESSIONAL EXPERIENCE

Software Engineer Intern - Jio Robotics

Sep 2025 - Jan 2026

- Designed scalable backend services and enterprise APIs supporting real-time operational workflows; optimized SQL queries and API execution paths, cutting response latency by 25%.
- Built modular backend components following clean architecture; participated in Agile/Scrum, peer code reviews, and CI/CD deployment workflows.

Research Analyst (Software Systems) - Central Institute of Tool Design

Aug 2023 - Jun 2024

- Designed an embedded system using Arduino and ultrasonic sensors for real-time object distance detection; implemented signal-processing logic and visualization workflows.

EDUCATION

M.S. Computer Science - University of Central Missouri | *May 2026*

CERTIFICATIONS

- [Building with the Claude API](#) - Anthropic · [Claude Code in Action](#) - Anthropic
- [AWS Cloud Architecting](#) - AWS Academy · [AWS Cloud Security Foundations](#) - AWS Academy
- [Advanced Software Engineering Simulation](#) - Walmart USA · [Software Engineering Simulation](#) - Skyscanner
- NPTEL - DBMS, Python, Embedded Systems · 200+ LeetCode problems (graphs, DP, recursion, sliding window) [Profile](#)